MIL-G-23010A (Wep)
13 March 1964
SUPERSEDING
MIL-G-23010 (Wep)
21 August 1961

MILITARY SPECIFICATION

GUN MECHANISM, 20MM, MARK 11 MOD 5

(ASSEMBLY)

1. SCOPE

1.1 Scope. This specification covers Gun Mechanism, 20MM, Mark 11 Mod 5, referred to herein as the gun mechanism.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-143	Specifications and Standards, Order of Precedence for Selection of.

DRAWINGS

Bureau of Naval Weapons
(Code Ident 10001)

LD 416843	Container, Shipping and Storage, for 20MM Aircraft Gun Mechanism Mark 11 Mod 5.
LD 416844	Container, Shipping and Storage, for 20MM Gun Barrels Mark 19 Mod 1 and Mark 20 Mod 1.
LD 522057	Pod, Gun, Mark 4 Mod 0 (Assembly).
LD 522078	Gun Mechanism, 20MM, Mark 11 Mod 5 (Assembly).

FSC 1005

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LD 522079	Loader, Gun, 20MM, Mark 2 Mod 1.
LD 615425	Link, Ammunition, 20MM Mk 6 Mod 4.
LD 615432	Kit, Gun Pod Accessories.
982517	High Pressure Test Rounds, 20MM, Mark 101 Mod 0.
982519	Target Practice Rounds, 20MM, Mark 105 Mod 0.
1445196	Dummy Round, 20MM, Mark 104 Mod 0.
1874673	Barrel Test Mount (for 20MM Gun Barrels, Mark 19 and Mark 20).

PUBLICATIONS

Military

Military Handbook H1	Cross-Index of Chemically Equivalent Specifications and Identification Code.

Bureau of Naval Weapons
(Code Ident 10001)

OP 2719	Gun Pod Mark 4 Mod 0 (with 20MM Aircraft Gun Mark 11 Mod 5), Description, Operation, and Maintenance.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of gun mechanisms shall be delivered for testing at a facility designated in the contract or order (see 6.2). The preproduction sample shall be manufactured using the methods proposed for production. Any production by the supplier prior to approval of the preproduction sample shall be at the supplier's risk.

3.2 Compliance with documents. Unless otherwise specified, the gun mechanism shall be in accordance with the requirements specified herein and in the applicable documents listed in section 2.

3.3 Materials. Materials used in the manufacture of the gun mechanism shall conform to the material specifications

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in the applicable documents listed in section 2. Subject to the approval of the Government representative, alternate materials may be selected in accordance with Military Handbook H1 in the order of precedence specified in Standard MIL-STD-143.

3.4 Design. The design of the gun mechanism shall be in accordance with the requirements specified herein and in the applicable documents listed in section 2. The gun mechanism shall be assembled with the other components of the Mark 4 Mod 0 Gun Pod and the gun pod shall be prepared for operation as specified in Publication OP 2719. The gun mechanism shall meet the requirements of 3.4.1 through 3.4.5.

3.4.1 Safety. When the switch controlling the gun pod clearing function is actuated, the gun mechanism shall cycle not fewer than three times in 500 milliseconds, on the first cycle ejecting cartridges or cases at the revolver eject position into the links in the loader. On subsequent cycles, the remaining cases or rounds shall be ejected from the revolver.

3.4.2 Function. When the switch controlling the gun pod charging function is actuated, the gun mechanism shall cycle, carrying rounds from the ram position to firing position. Any cartridges or cartridge cases at the eject position shall be ejected with a velocity sufficient to clear incoming links. When the gun pod trigger is actuated, the gun mechanism shall cycle, fire, and eject service rounds and cases at the rate of 4000 rounds per minute plus or minus 200 rounds per minute.

3.4.3 Life.

3.4.3.1 Operational life. The gun mechanism shall be capable of cycling, firing, and ejecting 130,000 service rounds and empty cases with normal parts replacement and servicing. See Publication OP 2719 for parts replacement schedule.

3.4.3.2 Shock and vibration. When preserved and packaged (see section 5), the gun mechanism shall withstand shocks, impacts from accidental dropping, and vibrations of logistic handling and transportation without damage.

3.4.3.3 Storage life. When preserved and packaged, the gun mechanism shall be capable of storage without deterioration for not less than three years.

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3.4.4 Interchangeability. Gun mechanism parts shall be capable of assembly without selection or fitting at industrial facilities and at the fleet level.

3.4.5 Coordination.

3.4.5.1 Service ammunition. The gun mechanism shall cycle, fire, and eject service ammunition and cartridge cases in the Mark 100 series.

3.4.5.2 Pod assembly. The gun mechanism shall assemble with the Pod (for Gun Pod Mark 4 Mod 0) without selection or fitting.

3.4.5.3 Loader. The gun mechanism shall assemble with the 20MM Gun Loader Mark 2 Mod 1 without selection or fitting.

3.4.5.4 Components. The pneumatic and electrical components of the gun mechanism shall be capable of nonselective connection with those of the Pod Assembly.

3.5 Construction. When assembled with the Government-loaned property of 3.9, the gun mechanism shall withstand the firing of nine High Pressure Test Rounds, 20MM, Mark 101 Mod 0.

3.6 Performance characteristics. When tested as specified in 4.6.3.3, the gun mechanism shall withstand the firing of the complete contents of four ammunition magazines without a stoppage, and shall meet the requirements of 3.4.2.

3.7 Details of components.

3.7.1 Gun barrels. Each Mark 19 Mod 2 gun barrel and each Mark 20 Mod 2 gun barrel intended for delivery to the Government shall withstand the firing of one Target Practice Round, 20MM, Mark 105 Mod 0 and of one High Pressure Test Round, 20MM, Mark 101 Mod 0. After firing, the increase in bore diameter, measured across the lands, shall not exceed 0.004 inch.

3.7.2 Electrical system. There shall be no induced voltage in the electrical system at any time which could energize firing circuits and accidentally fire ammunition.

3.8 Government-furnished property. The following, in quantities specified in the contract or order, are required

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to determine compliance of the gun mechanism with the requirements of this specification:

(a) Target Practice Rounds, 20MM, Mark 105 Mod 0.
(b) High Pressure Test Rounds, 20MM, Mark 101 Mod 0.
(c) Dummy Rounds, 20MM, Mark 104 Mod 0.
(d) Ammunition Links, Mark 6 Mods 4, 5, and 6.

3.9 Government-loaned property. The following are required to test the gun mechanism for conformance to the requirements of this specification:

(a) Loader, Gun, 20MM, Mark 2 Mod 1.
(b) Pod (for Gun Pod Mark 4 Mod 0).
(c) Ejector Rack Assembly, Aero 7A-1.
(d) Barrel Test Mount (for 20MM Gun Barrels, Mark 19 and Mark 20).
(e) Kit, Gun Pod Accessories.

3.10 Temperature extremes. The gun mechanism shall meet the requirements of 3.4.2 at any temperature between minus 65 degrees Fahrenheit (F.) and plus 165 degrees F.

3.11 Workmanship. The quality of workmanship on gun mechanism components and on the assembled gun mechanism shall assure conformance to all of the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the gun mechanism shall be classified as follows:

(a) Preproduction tests
(b) Acceptance tests.

4.3 Lot. Lot definition, size, and formation shall be as specified in Standard MIL-STD-105.

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4.4 Sampling. Unless otherwise specified herein, the provisions set forth in Standard MIL-STD-105 shall govern the establishment of sampling plans and procedures for inspection by attributes.

4.5 Examination.

4.5.1 Components. Prior to assembly of the gun mechanism, it shall be ascertained that all components procured under separate documents have been inspected, tested, and accepted in accordance with their respective documents.

4.5.2 Packing and marking. It shall be ascertained that packing and marking of the gun mechanism conform to section 5 of this specification.

4.6 Tests.

4.6.1 Acceptance tests. Sample gun mechanisms, selected in accordance with 4.4, shall be subjected to the acceptance tests specified herein, except that the tests of 4.6.1.2, 4.6.1.3, and 4.6.1.4 shall be performed on each gun mechanism and on each gun barrel intended for delivery to the Government. Lot quality shall be as specified in 4.6.2.

4.6.1.1 Visual and mechanical inspection. The mechanism shall be tested for conformance to 3.2 and 3.11.

4.6.1.2 Clearing test. The gun mechanism and loader shall be installed in the gun pod as specified in Publication OP 2719. The switch controlling the clearing function shall be actuated. The gun mechanism shall conform to 3.4.1.

4.6.1.3 Function tests. The gun pod components shall be assembled and prepared for operation as specified in Publication OP 2719, except that the magazine shall be loaded with two belts, each containing 130 links. Each belt shall be composed of one empty Mod 5 leading link, followed by 128 Mod 4 links and one Mod 6 trailing link. The belt intended to be fired through the last fire barrel shall contain target practice rounds. The other belt, beginning approximately at the thirtieth link, shall contain five successive dummy rounds. All other rounds in the belt shall be target practice rounds. Cycle the gun mechanism and fire the gun pod in three bursts of approximately 86 rounds per burst. Verify at the same time that the gun is charging automatically each time a dummy round is encountered. The gun mechanism shall meet the requirements of 3.4.2 and 3.4.5.1.

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4.6.1.4 Gun barrel proof test. Each Mark 19 and Mark 20 gun barrel shall be installed in the barrel test mount (see 3.9) and shall be tested by firing one target practice round and one high pressure test round. The gun barrels shall meet the requirements of 3.7.1.

4.6.2 Acceptance criteria. Lot quality for the test of 4.6.1.1 shall be judged in accordance with Standard MIL-STD-105, Acceptable Quality Level (AQL) 1.0 percent defective. Gun barrels failing to meet the requirements of 3.7.1 are defective and shall be removed from the lot. Failure of the gun mechanism to meet the requirements of 3.4.1, 3.4.2, and 3.4.5.1 shall be cause for rejection. The gun mechanism will not be judged defective if it is verified to the satisfaction of the Government representative that a gun pod stoppage or other malfunction was caused by a gun pod component other than the gun mechanism under test.

4.6.3 Preproduction tests. The preproduction tests shall include all of the acceptance tests of 4.6.1.1 through 4.6.1.4 and the tests specified herein.

4.6.3.1 Temperature extremes test. Mark 6 ammunition links shall be belted into four belts of 25 links, each composed of one empty Mod 5 leading link, followed by 23 Mod 4 links and one Mod 6 trailing link. Mod 4 links and the trailing link shall be loaded with Mark 105 target practice rounds. Two belts shall be loaded into the ammunition magazine of a gun pod assembled and prepared for operation in accordance with Publication OP 2719. The assembled gun pod shall be temperature-conditioned at minus 65 degrees F. plus 10.0 degrees F. minus 5.0 degrees F. until all components have been thermally stabilized at the specified temperature. The rounds shall be fired in a single burst. After firing, any required gun pod maintenance shall be performed as specified in Publication OP 2719. The two remaining belts shall be loaded into the magazine, and the gun pod shall be temperature-conditioned at 165 degrees F. plus 5.0 degrees F. minus 10.0 degrees F. until all components have been thermally stabilized at the specified temperature. The rounds shall be fired in a single burst.

4.6.3.2 Endurance firing test. The gun pod components shall be assembled and prepared for operation as specified in Publication OP 2719. The ammunition magazine shall be completely filled with Mark 6 ammunition links containing Mark 105 target practice rounds. The gun mechanism shall

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be cycled and the rounds shall be fired in bursts of approximately 125 rounds per burst. After the first, second, fourth, and fifth bursts, the assembled gun pod shall be allowed to cool for 3.0 minutes plus or minus 15 seconds. After the third burst, the gun pod shall be allowed to cool until the entire mass of the gun pod has been stabilized at the ambient temperature of the test facility. After the contents of the ammunition magazine have been completely expended, the gun pod shall again be allowed to cool to the test facility ambient temperature. Any required gun pod maintenance shall be performed as specified in Publication OP 2719. The ammunition magazine shall again be refilled and the test shall be repeated three additional times with cooling times and maintenance as specified herein.

4.6.3.3 Proof firing test. Mark 6 ammunition links shall be belted into two belts of 10 links, each composed of one Mod 5 leading link, followed by eight Mod 4 links and one Mod 6 trailing link. Each leading link shall remain empty. The belt intended to be fired through the first fire barrel shall be loaded with five Mark 105 target practice rounds, followed by four Mark 101 high pressure test rounds. The belt intended to be fired through the last fire barrel shall be loaded with four target practice rounds, followed by five high pressure test rounds. The gun pod shall be prepared for operation as specified in Publication OP 2719. The gun mechanism shall be cycled and the belts shall be fired in a single burst.

4.6.4 Preproduction acceptance criteria. Gun barrels shall not be required to conform to 3.7.1 following the test of 4.6.3.3. Failure of the gun mechanism to meet any of the other requirements of this specification shall be cause for rejection of the preproduction sample. The gun mechanism will not be judged defective if it is verified to the satisfaction of the Government representative that a gun pod stoppage or other malfunction was caused by a gun pod component other than the gun mechanism under test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation application. The lubricating oil specified in Drawing 1998259, listed in LD 416843, shall

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serve as the preservative for the gun mechanism.

5.1.1.2 Unit packaging. Not applicable.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior containers. Containers for the gun mechanism shall be in accordance with Container, Shipping and Storage, for 20MM Aircraft Gun Mechanism Mk 11 Mod 5, shown in drawings listed in LD 416843. Containers for the gun barrels shall be in accordance with Container, Shipping and Storage, for 20MM Gun Barrels Mark 19 Mod 1 and Mark 20 Mod 1.

5.2.2 Level C.

5.2.2.1 Exterior containers. Gun mechanisms and gun barrels shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall conform to the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special marking. None, unless otherwise specified.

5.3.2 Normal marking. In addition to the marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The gun mechanism covered by this specification is intended to be assembled with the 20MM Gun Loader Mark 2 Mod 1 and installed in the Pod (for Gun Pod Mark 4 Mod 0).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of preproduction sample if one is specified (see 3.1).
(c) Facility designated to evaluate the preproduction sample.

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(d) Government-loaned property (see 3.9).
(e) Government-furnished property (see 3.8).
(f) Lot size (see 4.3).
(g) Procedure for resubmittal of rejected lots.
(h) Preservation, packaging, and packing levels.
(i) Special marking of shipping containers if required.
(j) Invocation of Specification MIL-Q-9858

6.3 System reliability. Extreme care must be exercised in establishing the production system and quality control practices for parts covered by this specification to assure an adequate overall gun system reliability. Each subsystem must have considerably higher individual reliability than the overall 7,000 rounds per stoppage needed for the Mark 4 Gun Pods of which this item is a part. These reliabilities cannot be assured by an after the fact inspection to an AQL and are therefore beyond the written requirement of the specification. However, it has been shown that high reliabilities can be achieved even though inspection AQL's are relatively low (ammunition, for example) if a good production system and control is followed. To illustrate the component requirement, the subsystems must perform to stoppage rates no greater than those listed below if the 7,000 rounds per stoppage reliability level of the Mark 4 Gun Pod is to be maintained.

Item	Stoppage/rounds fired
Ammunition	1/100,000
Link	1/100,000
Loader	1/20,000
Gun	1/20,000
Pod	1/50,000

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-G-23010A(Wep) GUN MECHANISM, 20MM, Mk 11 Mod 5

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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